

Creating Maven Ansible Web App:

Step 1: Create Maven Web Application

The project was created using the Maven webapp archetype. The following command generates the base project structure:

```
mvn archetype:generate \
  -DgroupId=com.example \
  -DartifactId=MyAnsibleWebApp03 \
  -DarchetypeArtifactId=maven-archetype-webapp \
  -DinteractiveMode=false
```

```
vboxuser@Ubuntu:~/Desktop/Devops/Lab3$ mvn archetype:generate -DgroupId=com.example -DartifactId=MavenAnsibleWebApp03 -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/vboxuser/Desktop/Devops/Lab3
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MavenAnsibleWebApp03
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/vboxuser/Desktop/Devops/Lab3/MavenAnsibleWebApp03
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.159 s
[INFO] Finished at: 2026-05-09T11:27:20+05:30
[INFO]
```

Figure 1: Maven project creation – BUILD SUCCESS

After creation, the initial project structure is verified using the tree command:

```
vboxuser@Ubuntu:~/Desktop/Devops/Lab3/MavenAnsibleWebApp03$ tree
```

```
├── pom.xml
└── src
    ├── main
    │   ├── resources
    │   └── webapp
    │       ├── index.jsp
    │       ├── WEB-INF
    │       └── web.xml
    └── test
```

```
5 directories, 3 files
vboxuser@Ubuntu:~/Desktop/Devops/Lab3/MavenAnsibleWebApp03$
```

Figure 2: Initial Maven project structure

Step 2: Configure pom.xml

The pom.xml file is updated to include the Servlet API dependency and the Maven WAR plugin for packaging the application as a WAR file.

```
<dependencies>
```

```
<dependency>
  <groupId>javax.servlet</groupId>
  <artifactId>javax.servlet-api</artifactId>
  <version>4.0.1</version>
  <scope>provided</scope>
</dependency>
</dependencies>

<build>
  <plugins>
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-war-plugin</artifactId>
      <version>3.4.0</version>
    </plugin>
  </plugins>
  <finalName>MyAnsibleWebApp03</finalName>
</build>
```

```
vboxuser@Ubuntu:~/Desktop/Devops/Lab3/MavenAnsibleWebApp03$ tree
```

```
.
├── pom.xml
├── src
│   └── main
│       ├── resources
│       └── webapp
│           ├── index.jsp
│           ├── WEB-INF
│           └── web.xml
```

Figure 3: Project structure after pom.xml configuration

Step 3: Create Ansible Configuration Files

An `ansible/` directory is created inside the project containing two files: `hosts.ini` and `playbook.yml`.

```
mkdir ansible && cd ansible
gedit hosts.ini
gedit playbook.yml
```

hosts.ini

```
[web]
localhost ansible_connection=local ansible_python_interpreter=/usr/bin/python3
```

```

vboxuser@Ubuntu:~/Desktop/Devops/Lab3/MavenAnsibleWebApp03$ ls
pom.xml  src
vboxuser@Ubuntu:~/Desktop/Devops/Lab3/MavenAnsibleWebApp03$ mkdir ansible
vboxuser@Ubuntu:~/Desktop/Devops/Lab3/MavenAnsibleWebApp03$ cd ansible
vboxuser@Ubuntu:~/Desktop/Devops/Lab3/MavenAnsibleWebApp03/ansible$
vboxuser@Ubuntu:~/Desktop/Devops/Lab3/MavenAnsibleWebApp03/ansible$ gedit hosts.ini
vboxuser@Ubuntu:~/Desktop/Devops/Lab3/MavenAnsibleWebApp03/ansible$ gedit playbook.yml
vboxuser@Ubuntu:~/Desktop/Devops/Lab3/MavenAnsibleWebApp03/ansible$ ls
hosts.ini  playbook.yml
vboxuser@Ubuntu:~/Desktop/Devops/Lab3/MavenAnsibleWebApp03/ansible$ cd ..
vboxuser@Ubuntu:~/Desktop/Devops/Lab3/MavenAnsibleWebApp03$ tree
.
├── ansible
│   ├── hosts.ini
│   └── playbook.yml
├── pom.xml
└── src
    └── main
        ├── resources
        └── webapp
            ├── index.jsp
            └── WEB-INF
                └── web.xml

6 directories, 5 files
vboxuser@Ubuntu:~/Desktop/Devops/Lab3/MavenAnsibleWebApp03$

```

Figure 4: Creating ansible folder and files; project structure with ansible/

Step 4: Create Jenkinsfile

The Jenkinsfile defines the CI/CD pipeline with four stages: Checkout, Build, Archive, and Deploy. The Deploy stage triggers the Ansible playbook.

```

Open  [icon] Jenkinsfile
~/Desktop/Devops/Lab3/MavenAnsibleWebApp1

1 pipeline {
2   agent any // Use any available agent
3
4   environment {
5     LANG = 'en_US.UTF-8'
6     LC_ALL = 'en_US.UTF-8'
7   } // this has to be added only if you get an error saying UTF required is 8 but showing in ISO00009
8
9   tools {
10    maven 'Maven' // Ensure this matches the name configured in Jenkins
11  }
12  stages {
13    stage('Checkout') {
14      steps {
15        git branch: 'master', url: 'https://github.com/Someesvaar/MavenAnsibleWebApp1.git'
16      }
17    }
18
19    stage('Build') {
20      steps {
21        sh 'mvn clean package' // Run Maven build
22      }
23    }
24
25    stage('Archive') {
26      steps {
27        archiveArtifacts artifacts: 'target/*.war', fingerprint:true
28      }
29    }
30    stage('Deploy') {
31      steps {
32        sh 'mvn clean package'
33        sh 'ansible-playbook ansible/playbook.yml -i ansible/hosts.ini'
34      }
35    }
36
37  }
38 }
39
40 }

```

Figure 5: Jenkinsfile with pipeline stages (Checkout, Build, Archive, Deploy)

Step 5: Push Project to GitHub

The project is initialized as a Git repository and pushed to GitHub to serve as the SCM source for Jenkins.

```
git init
git add .
git commit -m "Initial Commit"
git remote add origin https://<api>@github.com/Someesvaar/MyAnsibleWebApp03.git
git push -u origin master
```

Step 6: Create Jenkins Pipeline Job

A new Pipeline project is created in Jenkins with the name MyAnsibleWebApp03-CICD. The pipeline definition is sourced from the SCM (GitHub repository).

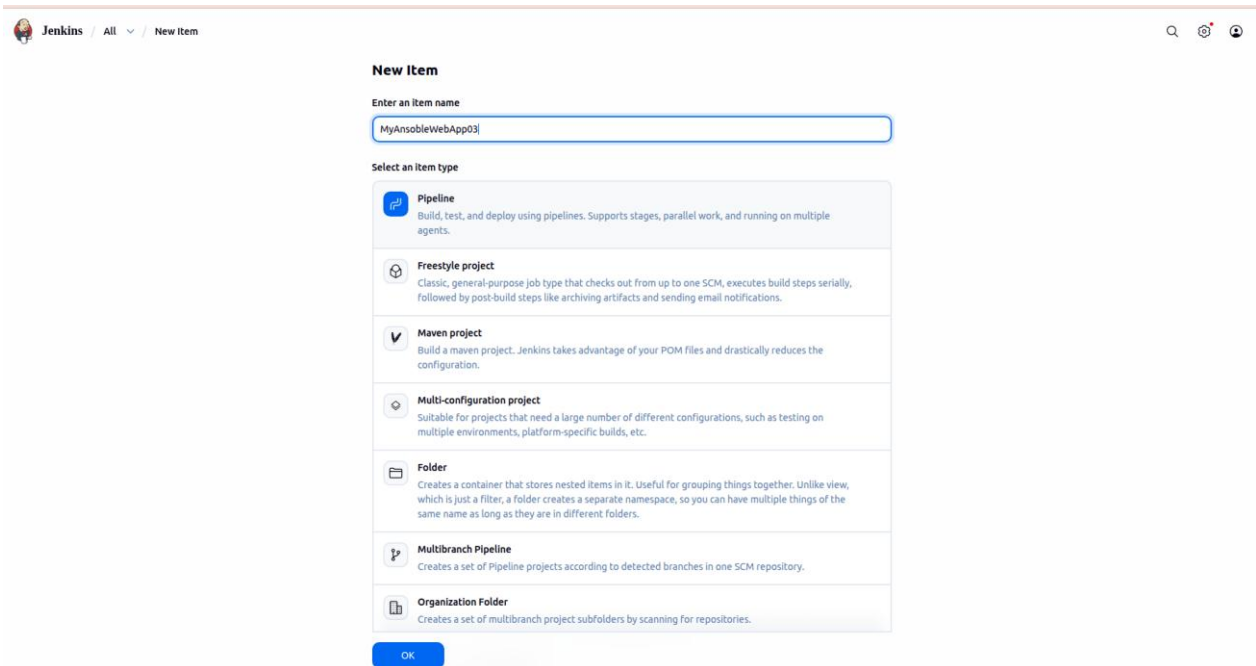


Figure 6: Jenkins New Item – creating Pipeline project

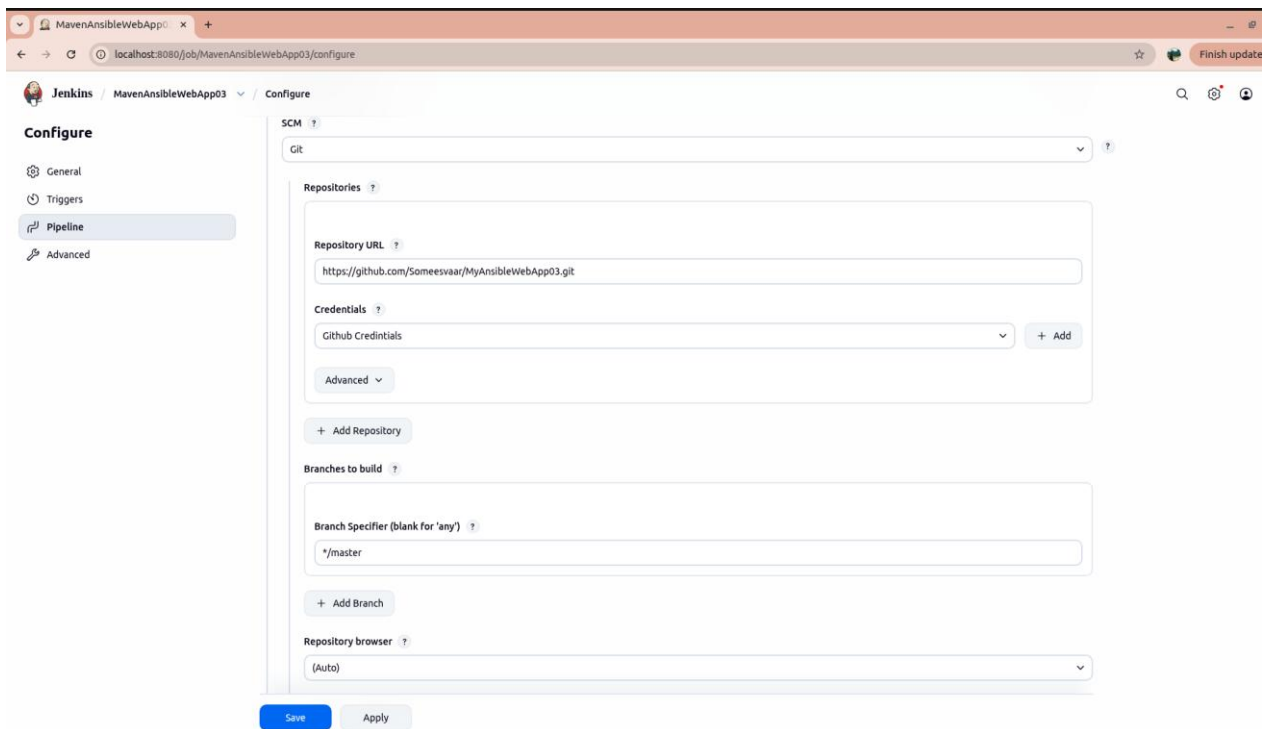


Figure 7: Jenkins Pipeline configured with GitHub SCM and credentials

Step 7: Update playbook.yml

After the Jenkins pipeline is set up, the Ansible playbook is updated with tasks to copy the generated WAR file to Tomcat's webapps directory and restart the server.

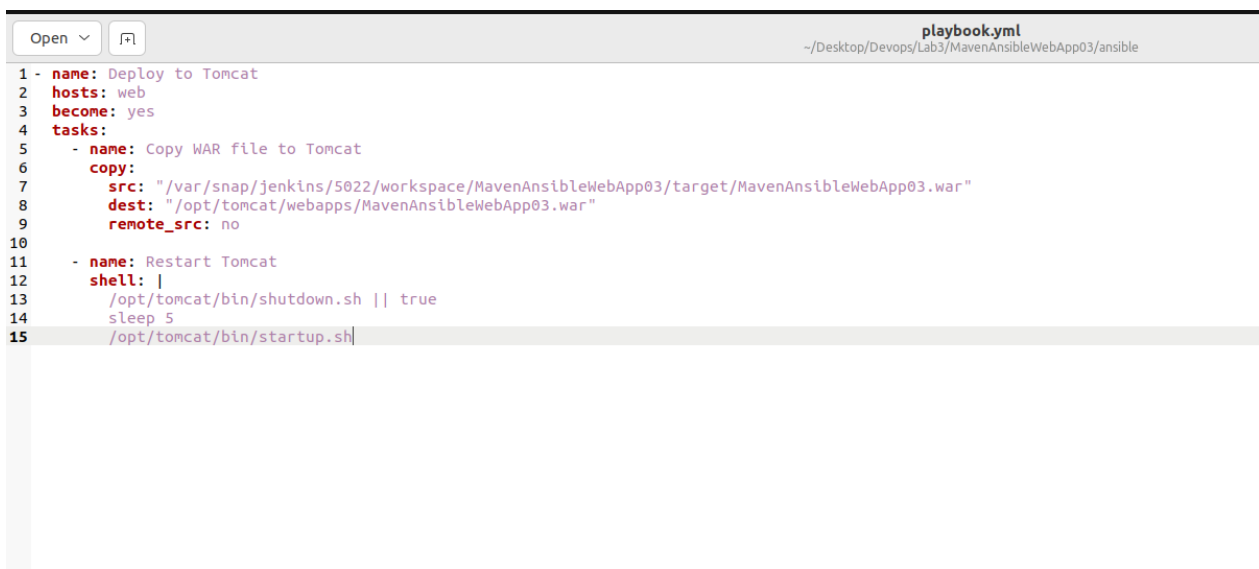


Figure 8: playbook.yml – Copy WAR to Tomcat and restart server

Step 8: Build Maven Project

The Maven build is executed locally to compile the code, generate the WAR file, and store it in the target/ directory.

```
mvn clean install
```

```

vboxuser@Ubuntu:~/Desktop/Devops/Lab3/HavenAnsibleWebApp03$ mvn clean install
[INFO] Scanning for projects...
[INFO] -----< com.example:HavenAnsibleWebApp03 >-----
[INFO] Building HavenAnsibleWebApp03 Maven Webapp 1.0-SNAPSHOT
[INFO] -----[ war ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ HavenAnsibleWebApp03 ---
[INFO] Deleting /home/vboxuser/Desktop/Devops/Lab3/HavenAnsibleWebApp03/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ HavenAnsibleWebApp03 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ HavenAnsibleWebApp03 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ HavenAnsibleWebApp03 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vboxuser/Desktop/Devops/Lab3/HavenAnsibleWebApp03/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ HavenAnsibleWebApp03 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ HavenAnsibleWebApp03 ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.5.1:war (default-war) @ HavenAnsibleWebApp03 ---
[INFO] Packaging webapp
[INFO] Assembling webapp [HavenAnsibleWebApp03] in [/home/vboxuser/Desktop/Devops/Lab3/HavenAnsibleWebApp03/target/HavenAnsibleWebApp03]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/vboxuser/Desktop/Devops/Lab3/HavenAnsibleWebApp03/src/main/webapp]
[INFO] Building war: /home/vboxuser/Desktop/Devops/Lab3/HavenAnsibleWebApp03/target/HavenAnsibleWebApp03.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ HavenAnsibleWebApp03 ---
[INFO] Installing /home/vboxuser/Desktop/Devops/Lab3/HavenAnsibleWebApp03/target/HavenAnsibleWebApp03.war to /home/vboxuser/.m2/repository/com/example/HavenAnsibleWebApp03/1.0-SNAPSHOT/HavenAnsibleWebApp03-1.0-SNAPSHOT.war
[INFO] Installing /home/vboxuser/Desktop/Devops/Lab3/HavenAnsibleWebApp03/pom.xml to /home/vboxuser/.m2/repository/com/example/HavenAnsibleWebApp03/1.0-SNAPSHOT/HavenAnsibleWebApp03-1.0-SNAPSHOT.pom
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] Total time: 3.600 s
[INFO] Finished at: 2026-05-09T11:47:18+05:30
[INFO] -----

```

Figure 9: mvn clean install – BUILD SUCCESS and WAR file generated

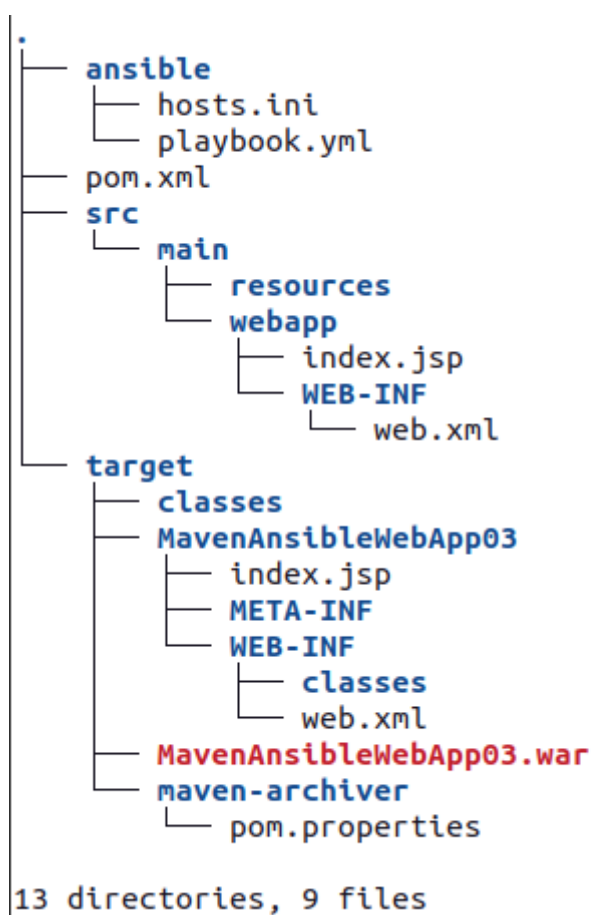


Figure 10: Final project structure showing target/MavenAnsibleWebApp03.war

Step 9: Run Jenkins Pipeline

The pipeline is triggered by clicking Build Now from the Jenkins dashboard. The pipeline runs through all stages: Checkout, Build, Archive, and Deploy.

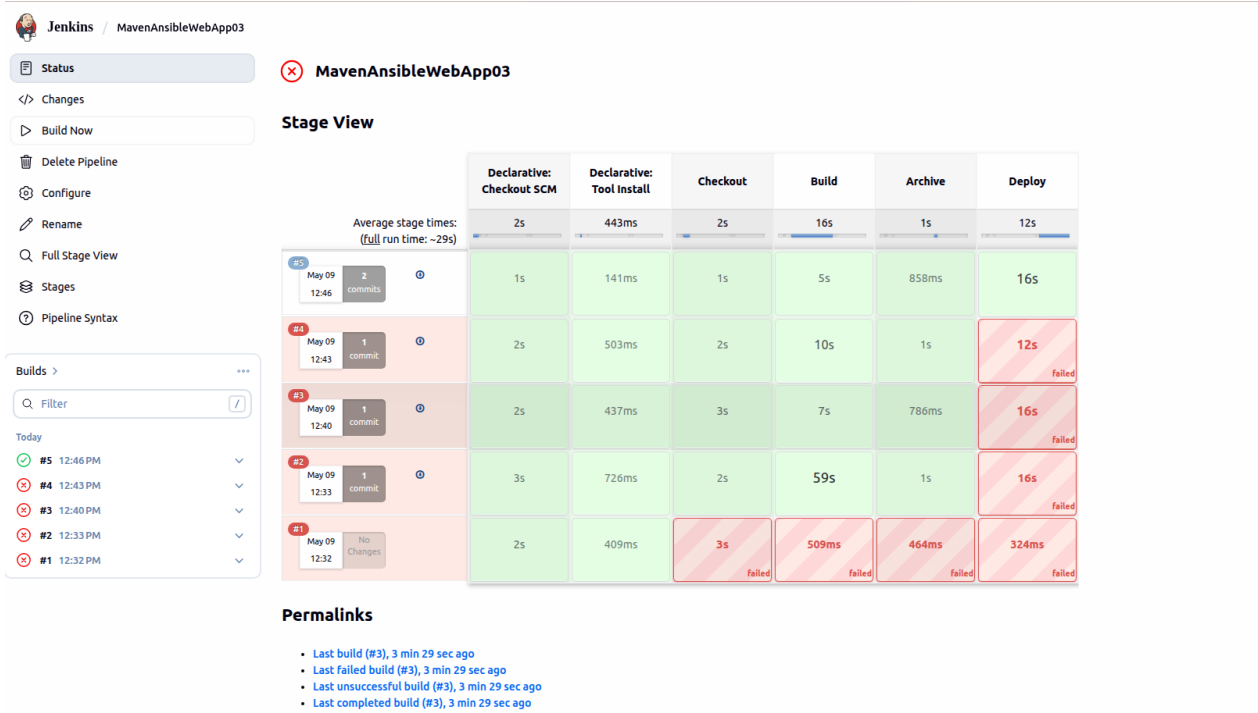


Figure 11: Jenkins Stage View – Build #5 completed successfully across all stages

Step 10: Verify Deployment on Tomcat

Tomcat is started and the deployed application is accessed via the browser at <http://localhost:9090/MavenAnsibleWebApp03>.

```
sudo -i
cd /opt/tomcat/bin
sh startup.sh
```

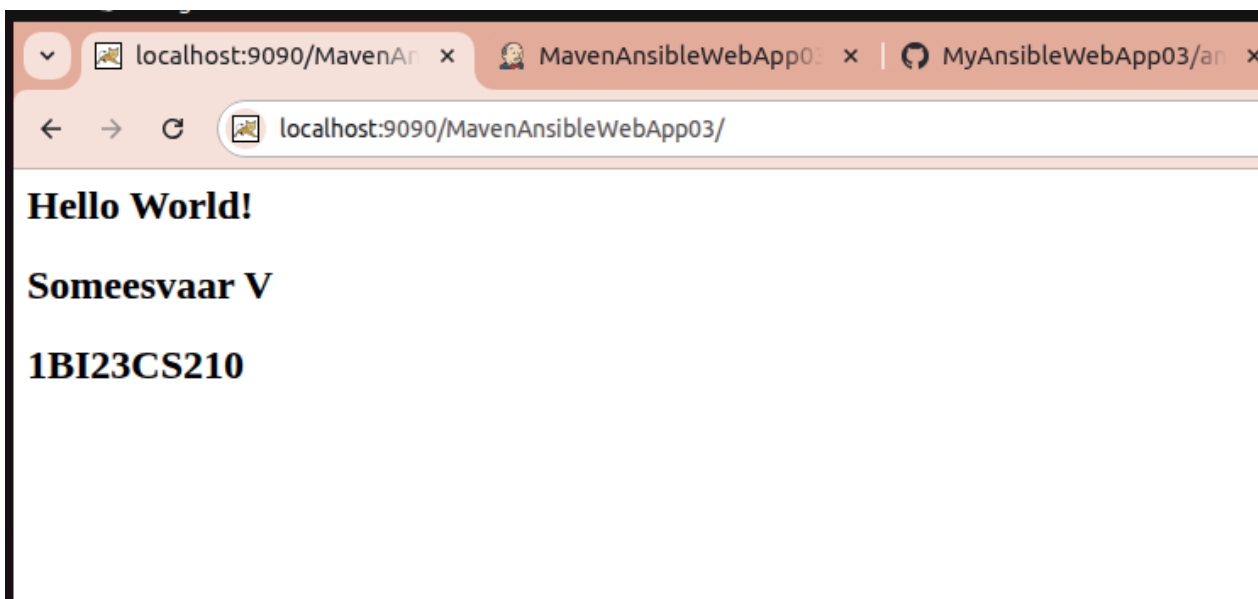


Figure 13: Application deployed and running – "Hello World! / Someesvaar V / 1BI23CS210"